

Importing the necessary libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Loading the dataset into a DataFrame

```
df = pd.read_csv("/content/nba-players.csv")
df.head()
```

	Unnamed: 0	name	gp	min	pts	fgm	fga	fg	3p_made	3pa	...	fta	ft	oreb	dreb	reb	ast	stl	blk	tov	target_5yrs
0	0	Brandon Ingram	36	27.4	7.4	2.6	7.6	34.7	0.5	2.1	...	2.3	69.9	0.7	3.4	4.1	1.9	0.4	0.4	1.3	0
1	1	Andrew Harrison	35	26.9	7.2	2.0	6.7	29.6	0.7	2.8	...	3.4	76.5	0.5	2.0	2.4	3.7	1.1	0.5	1.6	0
2	2	JaKarr Sampson	74	15.3	5.2	2.0	4.7	42.2	0.4	1.7	...	1.3	67.0	0.5	1.7	2.2	1.0	0.5	0.3	1.0	0
3	3	Malik Sealy	58	11.6	5.7	2.3	5.5	42.6	0.1	0.5	...	1.3	68.9	1.0	0.9	1.9	0.8	0.6	0.1	1.0	1
4	4	Matt Geiger	48	11.5	4.5	1.6	3.0	52.4	0.0	0.1	...	1.9	67.4	1.0	1.5	2.5	0.3	0.3	0.4	0.8	1

Explaining the column names.

Column Meaning

name - Player's name.

gP - Games Played during the rookie season.

min - Average Minutes Played per game.

pts- Average Points scored per game.

fgm -Average Field Goals Made per game.

fga -Average Field Goal Attempts per game.

3pMade - Average three-point field goals made per game.

3PA -Average three-point attempts per game.

ftm - Average Free Throws Made per game.

fta - Average Free Throw Attempts per game.

oreb - Average Offensive Rebounds per game.

dreb - Average Defensive Rebounds per game.

reb - Average Total Rebounds per game (Offensive + Defensive).

ast - Average Assists per game.

stl - Average Steals per game.

blk - Average Blocks per game.

tov - Average Turnovers per game.

target_5yrs - Target variable. Indicates whether the player remained in the NBA for at least five years. Typically: 1 = Yes, 0 = No

Defining the dependent variable y

```
y = df["target_5yrs"]
```

Dropping the index column

```
df.drop(columns=["Unnamed: 0"], inplace=True)
```

Defining the independent variables X

```
X = df.drop(columns=["name", "target_5yrs"])
```

Computing the correlation matrix

```
corr_matrix = X.corr()
print(corr_matrix)
```

	gp	min	pts	fgm	fga	fg	3p_made	\
gp	1.000000	0.590240	0.538471	0.542724	0.516625	0.296289	0.107423	
min	0.590240	1.000000	0.911822	0.903060	0.910247	0.203901	0.389920	
pts	0.538471	0.911822	1.000000	0.990834	0.979733	0.255333	0.346682	
fgm	0.542724	0.903060	0.990834	1.000000	0.980050	0.291693	0.289007	
fga	0.516625	0.910247	0.979733	0.980050	1.000000	0.129798	0.390253	
fg	0.296289	0.203901	0.255333	0.291693	0.129798	1.000000	-0.294471	
3p_made	0.107423	0.389920	0.346682	0.289007	0.390253	-0.294471	1.000000	
3pa	0.098772	0.403258	0.356751	0.299057	0.413560	-0.350658	0.982616	
3p	0.037133	0.168070	0.154955	0.122542	0.201186	-0.336502	0.591733	
ftm	0.482123	0.791000	0.896297	0.848019	0.826616	0.245776	0.158472	
fta	0.479487	0.779609	0.880703	0.840408	0.805559	0.300154	0.095396	
ft	0.196299	0.239878	0.258931	0.223566	0.269614	-0.161183	0.314355	
oreb	0.401136	0.573062	0.575106	0.596687	0.504212	0.511367	-0.219010	
dreb	0.466840	0.745513	0.693934	0.703278	0.640123	0.410555	0.016570	
reb	0.460406	0.709707	0.676849	0.691186	0.614328	0.465423	-0.072503	
ast	0.372749	0.629015	0.552338	0.532534	0.589818	-0.108797	0.376604	
stl	0.451137	0.757034	0.675341	0.662640	0.690168	0.056658	0.306908	
blk	0.276498	0.399088	0.387043	0.398125	0.322184	0.391626	-0.158535	
tov	0.518167	0.826500	0.850366	0.834352	0.845989	0.121806	0.258369	

	3pa	3p	ftm	fta	ft	oreb	dreb	\
gp	0.098772	0.037133	0.482123	0.479487	0.196299	0.401136	0.466840	
min	0.403258	0.168070	0.791000	0.779609	0.239878	0.573062	0.745513	
pts	0.356751	0.154955	0.896297	0.880703	0.258931	0.575106	0.693934	
fgm	0.299057	0.122542	0.848019	0.840408	0.223566	0.596687	0.703278	
fga	0.413560	0.201186	0.826616	0.805559	0.269614	0.504212	0.640123	
fg	-0.350658	-0.336502	0.245776	0.300154	-0.161183	0.511367	0.410555	
3p_made	0.982616	0.591733	0.158472	0.095396	0.314355	-0.219010	0.016570	
3pa	1.000000	0.584853	0.173533	0.108388	0.323612	-0.231897	0.011226	
3p	0.584853	1.000000	0.036245	-0.027249	0.332458	-0.289399	-0.125716	
ftm	0.173533	0.036245	1.000000	0.980505	0.257818	0.583865	0.653823	
fta	0.108388	-0.027249	0.980505	1.000000	0.114872	0.653445	0.700863	
ft	0.323612	0.332458	0.257818	0.114872	1.000000	-0.146786	-0.022905	
oreb	-0.231897	-0.289399	0.583865	0.653445	-0.146786	1.000000	0.838580	
dreb	0.011226	-0.125716	0.653823	0.700863	-0.022905	0.838580	1.000000	
reb	-0.080939	-0.193082	0.653833	0.711425	-0.071105	0.932694	0.978177	
ast	0.410531	0.266898	0.476214	0.428624	0.296315	-0.012109	0.186679	
stl	0.338631	0.198237	0.600158	0.580065	0.207205	0.286545	0.411894	
blk	-0.172150	-0.246358	0.407466	0.468974	-0.161152	0.648346	0.688135	
tov	0.283925	0.111495	0.804990	0.798936	0.199742	0.421695	0.570187	

	reb	ast	stl	blk	tov
gp	0.460406	0.372749	0.451137	0.276498	0.518167
min	0.709707	0.629015	0.757034	0.399088	0.826500
pts	0.676849	0.552338	0.675341	0.387043	0.850366
fgm	0.691186	0.532534	0.662640	0.398125	0.834352
fga	0.614328	0.589818	0.690168	0.322184	0.845989
fg	0.465423	-0.108797	0.056658	0.391626	0.121806
3p_made	-0.072503	0.376604	0.306908	-0.158535	0.258369
3pa	-0.080939	0.410531	0.338631	-0.172150	0.283925
3p	-0.193082	0.266898	0.198237	-0.246358	0.111495
ftm	0.653833	0.476214	0.600158	0.407466	0.804990
fta	0.711425	0.428624	0.580065	0.468974	0.798936
ft	-0.071105	0.296315	0.207205	-0.161152	0.199742
oreb	0.932694	-0.012109	0.286545	0.648346	0.421695
dreb	0.978177	0.186679	0.411894	0.688135	0.570187
reb	1.000000	0.119081	0.381154	0.699672	0.536716

```
upper = corr_matrix.where(
    np.triu(np.ones(corr_matrix.shape), k=1).astype(bool)
)
```

```
high_corr = [
```

```

        (column, row, upper.loc[row, column])
        for column in upper.columns
        for row in upper.index
        if pd.isna(upper.loc[row, column]) and abs(upper.loc[row, column]) > 0.95
    ]

    for feature1, feature2, corr in high_corr:
        print(f"{feature1} - {feature2}: {corr:.2f}")

```

```

fgm - pts: 0.99
fga - pts: 0.98
fga - fgm: 0.98
3pa - 3p_made: 0.98
fta - ftm: 0.98
reb - dreb: 0.98

```

```
X = X.drop(columns=["fgm", "fga", "3pa", "fta", "dreb"])
```

Detailed explanations of the columns that I removed and why I removed them

I removed index column because it is just a row number and should not be used as feature

I removed name column because contains no information about players characteristics or career outcome.

reb is the sum of oreb and dreb, and dreb and reb are highly correlated. Therefore I dropped dreb

Points per game(pts) is more interpretable and summarizes scoring better than fgm and fga. Therefore I dropped fgm and fga.

I dropped free throw attempt(fta) and kept free throw made(ftm) because free throws made is more informative than attempts only.

Start coding or [generate](#) with AI.

Engineering a new composite feature : Points Per Minute

```
df["Points_Per_Minute"] = df["pts"] / df["min"]
```

```

X["Points_Per_Minute"] = df["Points_Per_Minute"]
new_df = pd.concat([X, y], axis=1)

```

Handling null values

```
new_df.isnull().sum()
```

	0
gp	0
min	0
pts	0
fg	0
3p_made	0
3p	0
ftm	0
ft	0
oreb	0
reb	0
ast	0